

JOSE ANDRES BARRIENTOS

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PROFESSIONAL SUMMARY

Data engineer with 4+ years building production ETL pipelines and data warehouses using Python, SQL, Spark, and AWS. Delivered \$2M+ revenue impact through automated data workflows processing 10M+ records daily. Pursuing Master's in Software Engineering at UMD.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript

Data Engineering: ETL/ELT, Apache Airflow, dbt, Apache Spark/PySpark, Data Warehousing, Data Modeling (Star/Snowflake Schema)

Databases: PostgreSQL, MySQL, MongoDB, Redis, Query Optimization, Indexing, Performance Tuning

Cloud Platforms: AWS (S3, Glue, Redshift, Athena, Lambda, RDS), Docker, CI/CD, Git

Data Analysis & ML: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Statistical Analysis, Predictive Modeling

BI & Visualization: Power BI, Tableau, Plotly, Matplotlib, Excel (Advanced), Dashboard Design

EDUCATION

Master of Engineering in Software Engineering

Expected May 2027

University of Maryland, College Park | College Park, MD

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Distributed Systems, Database Systems, Software Engineering

Bachelor of Science in Industrial Engineering

December 2019

Universidad Dr. Jose Matias Delgado | San Salvador, El Salvador

PROFESSIONAL EXPERIENCE

Data Analyst | Intradeco

September 2021 – August 2025

San Salvador, El Salvador

- Designed and implemented automated ETL pipelines using Python and SQL processing 10M+ rows daily from multiple data sources (REST APIs, databases, CSV files), transforming raw data into analytics-ready datasets that enabled \$2M+ revenue impact
- Built data warehouse architecture with dimensional modeling (star schema) using PostgreSQL, optimizing query performance by 93% (45s to 3s) for high-traffic analytics workloads
- Developed interactive Power BI dashboards integrating SQL databases and REST APIs, providing real-time visibility into KPIs that accelerated category manager decision-making by 30%
- Automated weekly reporting workflows using Python and Excel VBA, eliminating 6 hours of manual data preparation per week (75% efficiency gain) and ensuring consistent data delivery to stakeholders

Quality Assurance Coordinator | SigmaQ Specialty

January 2021 – June 2021

San Salvador, El Salvador

- Designed comprehensive testing procedures and data collection frameworks ensuring product compliance with industry standards, reducing defect rates by 18%
- Analyzed quality metrics and customer feedback data using statistical methods to identify process improvement opportunities, implementing data-driven initiatives that increased operational efficiency by 12%

Customer Service Representative | Telus International

June 2020 – December 2020

San Salvador, El Salvador

- Resolved 40+ customer inquiries daily across multiple channels, maintaining 95%+ satisfaction rating through data-driven problem-solving and technical troubleshooting
- Analyzed customer complaint patterns to identify recurring issues, collaborating with technical teams to implement solutions that reduced support volume by 12%

TECHNICAL PROJECTS

GitHub Issues Analytics Platform | University of Maryland

August 2025 – December 2025

Python, Scikit-learn, Pandas, Plotly, Matplotlib | Team Project (Software Engineering)

- Built end-to-end data analytics pipeline processing 5,200+ GitHub issues in JSON format, implementing data extraction, transformation, and loading workflows with 90% test coverage
- Developed machine learning models (Random Forest, Gradient Boosting) for predictive analytics, achieving 80% accuracy in classifying issue priority and complexity for automated triage
- Created 7 interactive data visualizations using Plotly and Matplotlib including contributor lifecycle analysis, engagement heatmaps, and temporal activity trends
- Designed modular architecture enabling automated report generation with statistical analysis and predictive insights delivered to stakeholders

Road Damage Detection Using Deep Learning | University of Maryland

August 2025 – December 2025

Python, PyTorch, TensorFlow, Google Colab, NumPy, Pandas | Team Project (AI/ML Fundamentals)

- Engineered complete data pipeline for image classification processing 907 images through preprocessing, feature extraction, model training, and evaluation stages
- Implemented data augmentation strategies and transfer learning (ImageNet pretraining) improving validation accuracy by 16% and reducing overfitting by 12.5%
- Conducted comprehensive ablation studies analyzing impact of hyperparameters (batch size, learning rate, augmentation) on model performance, reducing training time by 44%
- Compared 5 neural network architectures (ResNet-18, MobileNetV3, EfficientNet-B0, ViT-Tiny) demonstrating 86% parameter reduction potential for edge deployment

AI Game Engine with Minimax Algorithm | University of Maryland

August 2025 – December 2025

Python, Algorithm Design, Game Theory | Course Project (Python for Engineering AI)

- Designed and implemented minimax algorithm with alpha-beta pruning for optimal decision-making, achieving 40-60% reduction in computational complexity
- Developed heuristic evaluation function analyzing positional advantages and winning patterns, enabling AI to achieve 90%+ win/draw rate
- Optimized recursive search algorithm with configurable depth parameter balancing computational efficiency (<1s per move at depth 6) with strategic decision quality